A Multivariate Analysis of the Relationship Between Demographic Characteristics and Social Values

STA437 Project 3

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Introduction

The World Values Survey (WVS) Wave 7 dataset provides a large cross-national sample of individuals' demographic characteristics and social values. In this project, the responses from 19,364 individuals across 10 countries will be examined to determine whether demographic variables such as age, gender, education, and income are systematically associated with the value variables, including religion, trust, life satisfaction, gender equality, and democracy. The central goal of the report is to understand the joint relationship between demographic and value variables with multivariate methods taught in this course.

This research is important because many social value variables are conceptually related and usually change together. For example, variables like trust in others, life satisfaction, and attitudes toward democracy often reflect an orientation toward social cohesion and institutional confidence. The multivariate approach enables these relationships to be analyzed simultaneously, so that shared variation would not be missed. In the meantime, although demographic variables are widely used as predictors in social science research, their explanatory power remains limited. As a result, this project asks not only if demographic variables matter but also how strong and structured their relationship with value variables is.

First, the analysis will summarize the dataset and address preprocessing decisions, particularly missingness. Second, exploratory analysis of correlation structure and multivariate assumption checks will be conducted. Then PCA would serve as a baseline method for assessing the data's dimensionality. Next, CCA would become the primary method for studying the relationships between demographic and value variables. Lastly, a multivariate linear regression is fit to evaluate predictive performance using cross-validation, while conducting a sensitivity analysis that removes the variables with the most problematic missingness.

Data Description and Preprocessing

There are 14 variables in total to focus on: four demographic variables (gender, age, education level, and income decile) and ten value variables (importance of religion, family, work, and friends, national pride, trust in people, life satisfaction, political orientation, gender equality, and importance of democracy). All negative WVS codes indicating missing or invalid responses were treated as missing values. A few variables were reverse-coded to ensure higher values consistently represent higher levels of the construct. This step was added because it made the relationships between variables easier to interpret while avoiding sign inconsistencies.

Table 1: Sample and preprocessing summary

	Value
Total observations (raw)	19364
Complete-case observations	11919
Countries (raw)	10
Countries (complete-case)	9
Max missing (%)	0.295445
Variable with max missing	political_orientation

Table 1 above summarizes the sample before and after preprocessing. After restricting to complete cases across all selected variables, the sample size decreases from 19,364 respondents across 10 countries to 11,919 respondents across 9 countries. This is a substantial reduction and highlights the major limitation that the availability of complete information can vary considerably across variables.

Figure 1. Missingness profile of WVS analysis variables

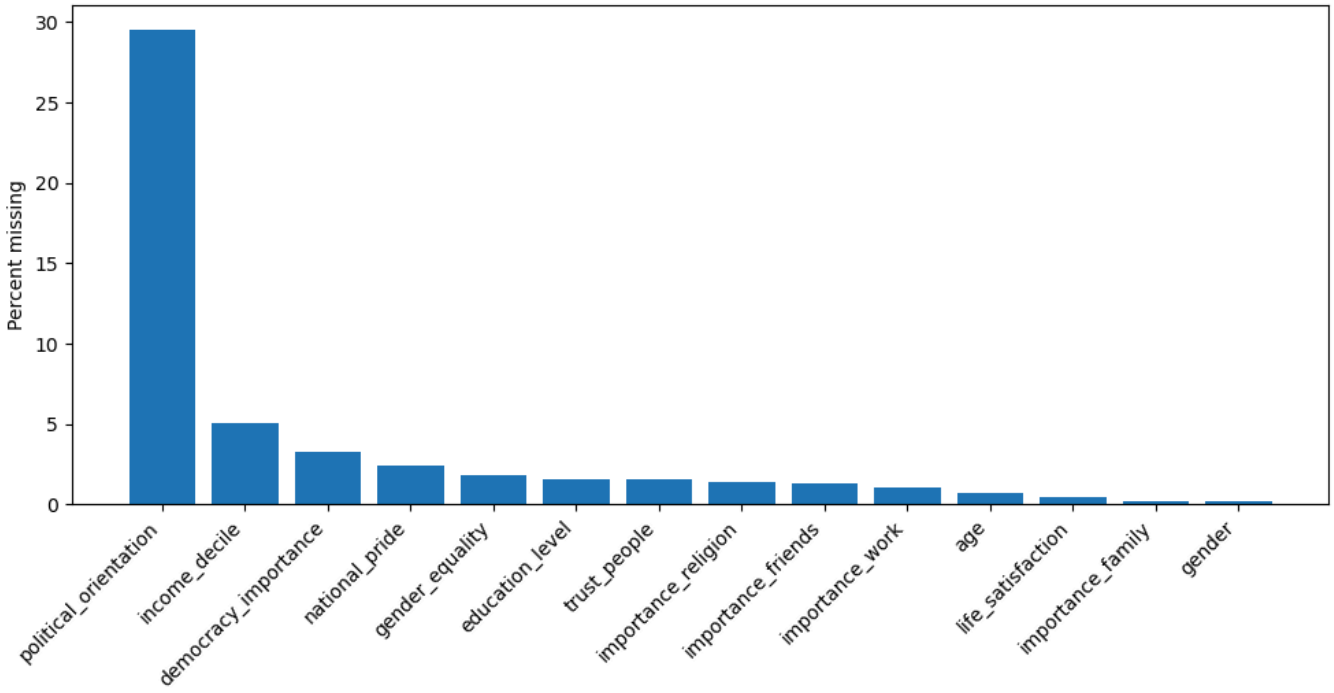
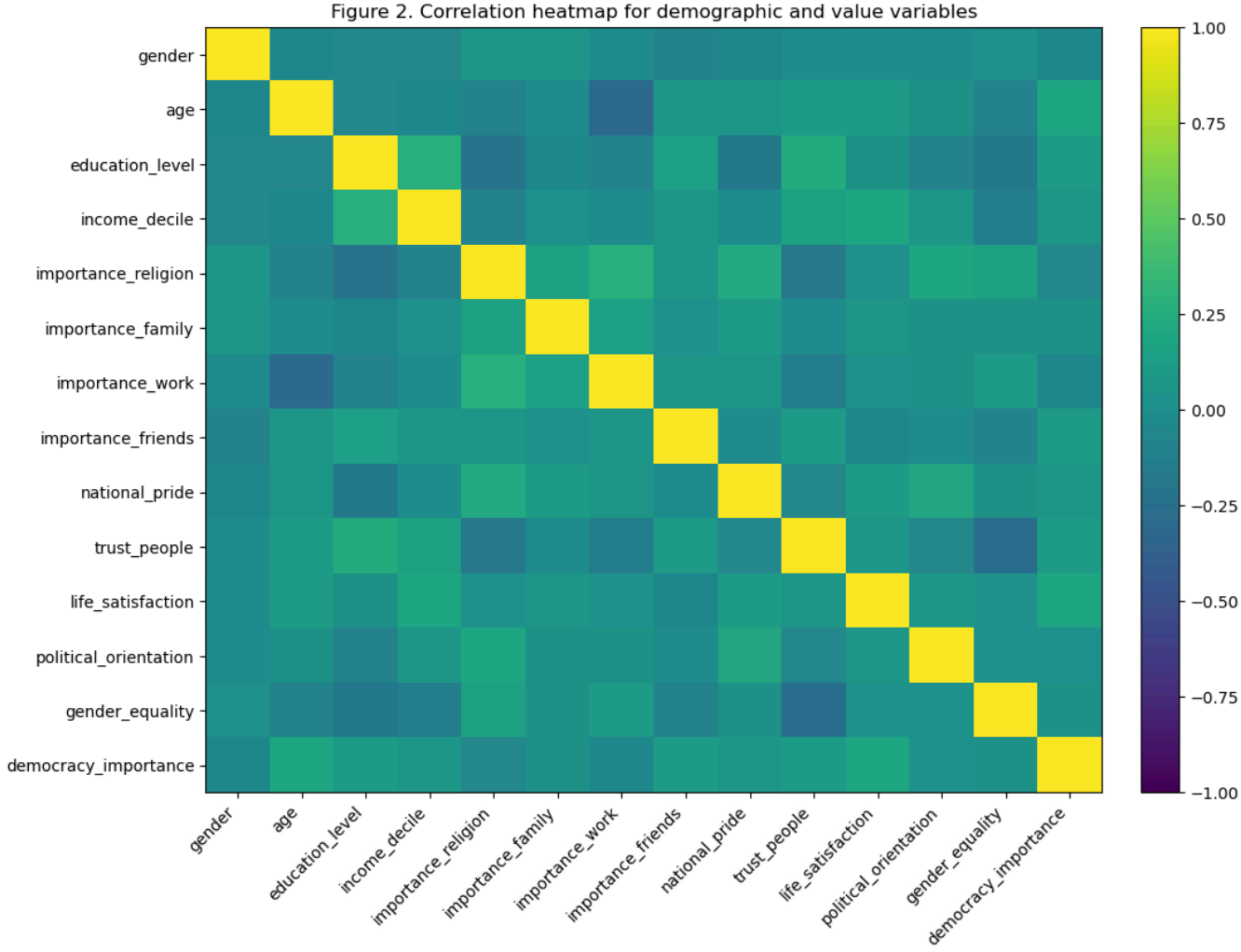


Figure 1 shows the distribution of missingness across variables. It is clear that missingness is uneven, with political orientation accounting for 30% of missingness and other variables showing low missingness. The importance of income decile and democracy shows moderate missingness, yet political orientation still significantly exceeds the remaining variables. This pattern suggests that the complete-case approach in the main analysis may result in a nontrivial reduction in sample size, affecting representativeness. In addition, the pattern motivates a sensitivity analysis in which political orientation is excluded to be conducted later. This would allow more observations to be retained and provide a check on the robustness of the results.

Exploratory Data Analysis

Below, the correlation heatmap of the analysis variables is used to verify that they exhibit a meaningful joint structure. Largely independent variables would make it clear that multivariate techniques add little value, but if the converse holds, the presence of correlations would justify methods that capture shared variation.

Figure 2. Correlation heatmap for demographic and value variables



The heatmap reveals moderate yet meaningful relationships across both demographic and value variables. For instance, education and income seem to be associated with several value outcomes, suggesting that socioeconomic position may be important in shaping values. Meanwhile, value variables such as trust, life satisfaction, gender equality, and democracy importance all exhibit internal correlations, meaning they are not independent. Nonetheless, the heatmap still does not display a single dominant block of correlations but rather a structure that appears diffuse and multidimensional. This observation suggests that relationships exist but are not attributable to a single underlying factor, consistent with the presumed complexity of social values, in which multiple effects contribute to observed responses.

Table 2: Bartlett's Test of Sphericity

	chi2	df	p_value
0	11991.7674	91.0	0.0

Table 3: Mardia Multivariate Normality Test

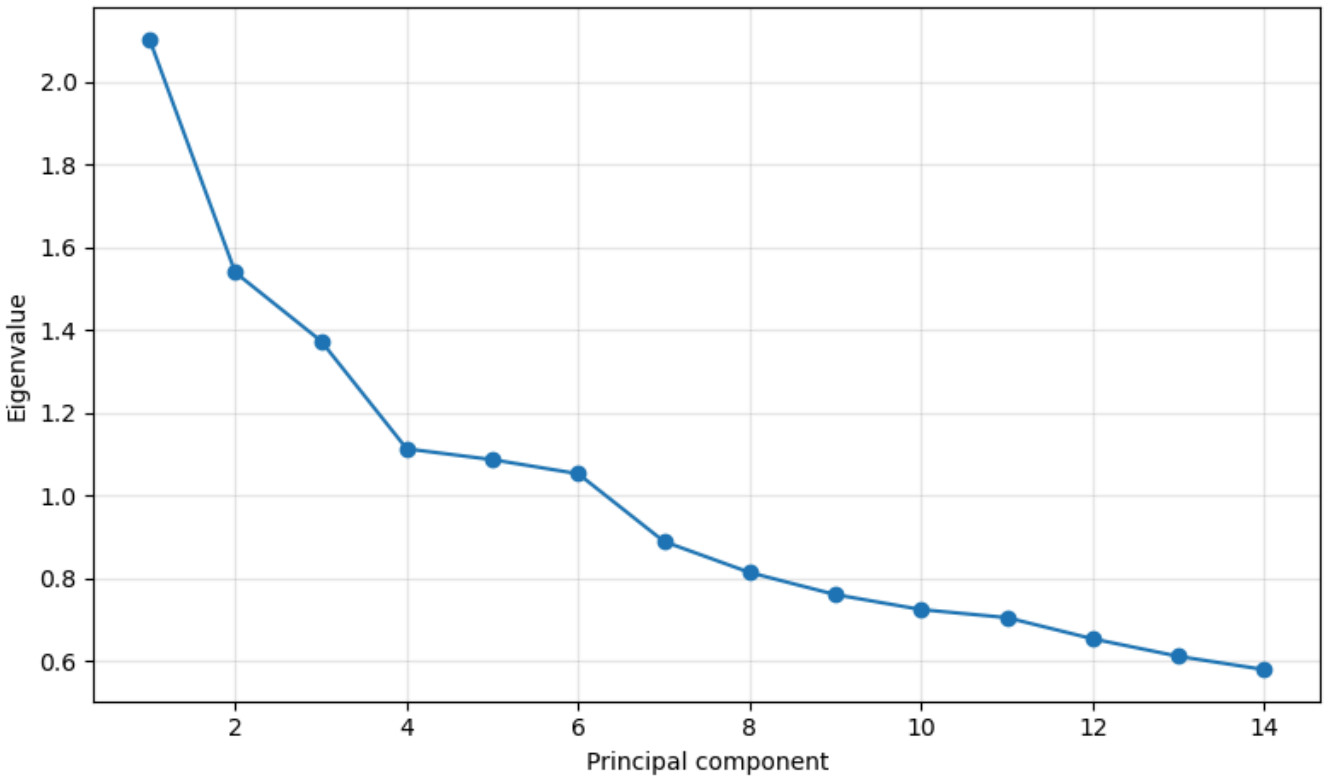
	n	p	mardia_skewness	skewness_p	mardia_kurtosis	kurtosis_p
0	11919	14	35.8902	0.0	248.8222	0.0

These observations are supported by the formal multivariate assumption tests. Bartlett's test of sphericity indicates a strong rejection of the null hypothesis that the correlation matrix is an identity matrix ($p < 0.001$), which confirms that multivariate analysis is justified due to the sufficient correlation of the variables. Additionally, Mardia's multivariate normality tests reject normality for both skewness and kurtosis, which is expected given the presence of skewness in religious importance and trust. This violates strict assumptions of some multivariate methods, yet never invalidates the analysis. Rather, it can be interpreted as a support for the use of multiple complementary approaches.

Baseline PCA

To examine the dataset's dimensionality, PCA is used as a baseline descriptive method. Although it does not directly account for the relationship between cemogtraphic and value variables, it could provide an answer to how many dimensions are needed to summarize the overall variation.

Figure 3. PCA scree plot for the complete-case analysis matrix



In the scree plot of eigenvalues above, the first principal component is the largest, but the decline in eigenvalues is not sharp but gradual. In addition, several components contribute meaningfully to the total variance, indicating that a single dimension cannot adequately capture the data. The relationships among variables are spread across multiple latent directions.

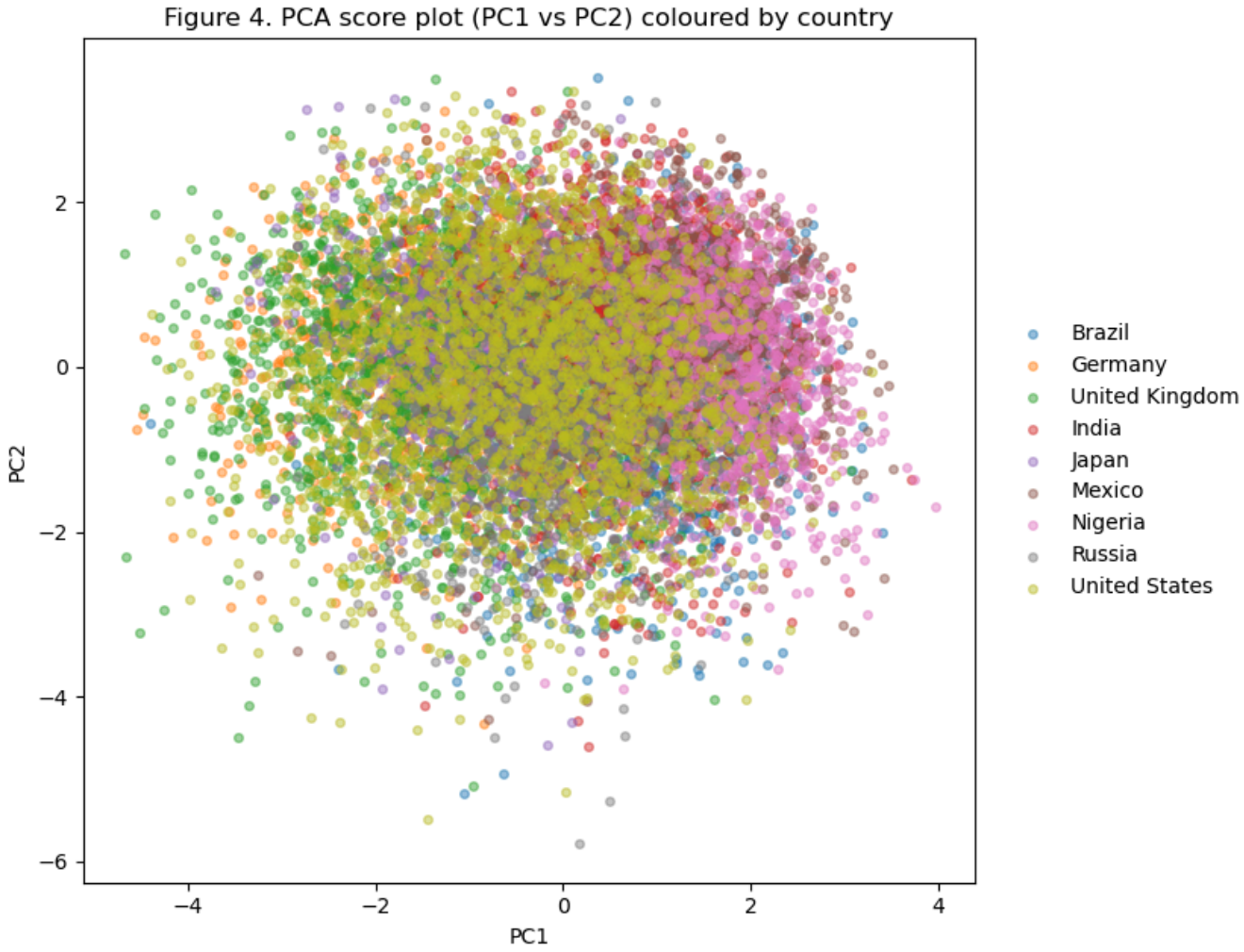


Figure 4 is the PCA score plot for the first two components, coloured by country. A substantial overlap exists across countries, with minor shifts in position, indicating that the principal components do not correspond to country-level clusters. Variation is continuous across respondents, regardless of country. This finding, in particular, provides additional evidence that the data structure is multidimensional and motivates the use of CCA, which can directly model relationships between predefined variable blocks.

Canonical Correlation Analysis

Canonical correlation analysis, or CCA, is the main method in the report for examining the relationship between demographic and value variables. CCA can identify the linear combinations of each block that are maximally correlated with one another, providing a more direct measure of the association between two sets.

`Table 4: Canonical Correlations`

	canonical_correlation
Function1	0.4779
Function2	0.2776
Function3	0.1866
Function4	0.1345

Table 5: CCA Redundancy Summary

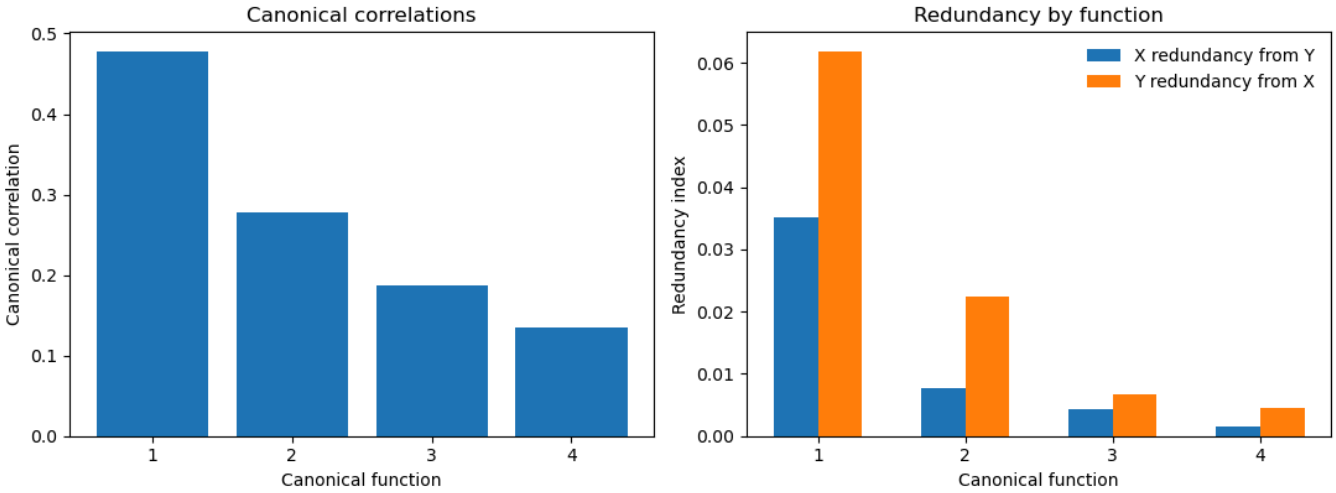
	corr	corr_sq	X_red_from_Y	Y_red_from_X
Function1	0.4779	0.2284	0.0352	0.0618
Function2	0.2776	0.0770	0.0076	0.0224
Function3	0.1866	0.0348	0.0043	0.0067
Function4	0.1345	0.0181	0.0016	0.0045

Table 6: CCA Significance Tests

	Canonical Correlation	Wilks' lambda	Num DF	Den DF	F Value	Pr > F
0	0.477863	0.674957	40	45121.473831	123.22079	0.0
1	0.27756	0.874697	27	34754.778765	60.37955	0.0
2	0.18662	0.947709	16	23802.0	40.490029	0.0
3	0.134516	0.981906	7	11902.0	31.332651	0.0

According to the results in the above tables, the first canonical correlation is 0.4779, and the subsequent correlations are 0.2776, 0.1866, and 0.1345, respectively. The first canonical function dominates in magnitude, despite all of them being statistically significant. Squaring the first canonical correlation reveals that 22.8% of the variance in the canonical varieties is shared. However, redundancy measures are way smaller, so that the proportion of variance in one block explained by the other is modest. This distinction is important in showing that the strong correlation in canonical variates does not imply high explanatory power for individual variables.

Figure 5. CCA strength summary



Further, Figure 5 illustrates the strength and redundancy of the canonical functions. Again, the first canonical function is substantially larger than the others, with redundancy decreasing rapidly in subsequent functions. This means that later canonical dimensions contribute very little to interpretation, and the first dimension accounts for most of the meaningful relationship between demographics and value.

Figure 6. First canonical variates from CCA

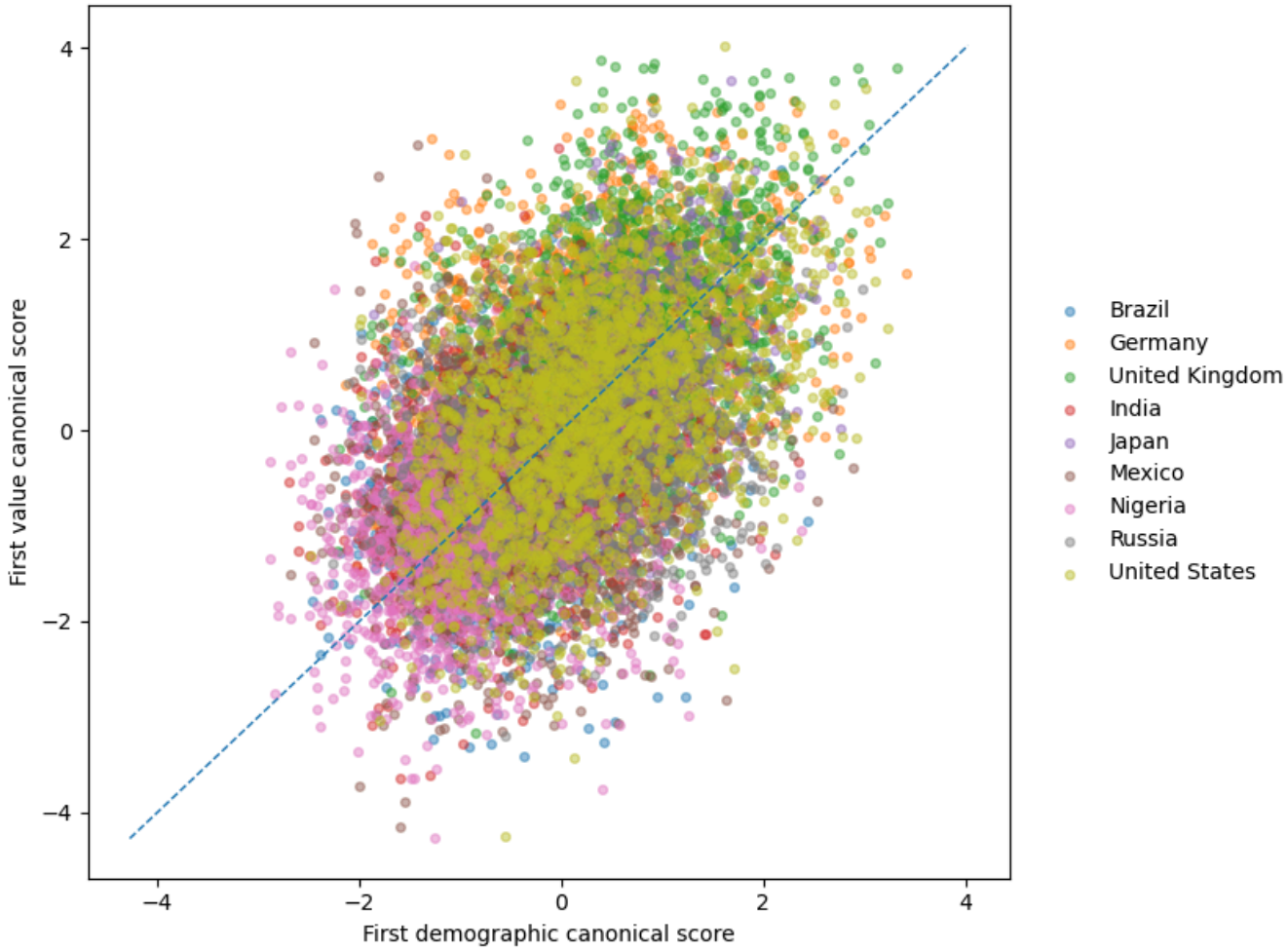


Figure 6 in particular exhibits the first pair of canonical variates. A positive association is obvious, further confirming the linkage between demographic and value variables, although the spread of observations indicates this is only a moderate relationship. The variability around the trend is considerable, and countries also overlap extensively, which reinforces the idea that demographic effects operate along continuous gradients.

MVLR, Validation, and Sensitivity Analysis

Multivariate linear regression models are proposed to assess the predictive power of demographic variables, thereby complementing the CCA results.

Table 7: MVLR Fit Statistics

	response	r_squared	adj_r_squared	f_p_value
2	importance_work	0.1007	0.1004	0.0
5	trust_people	0.0850	0.0847	0.0
0	importance_religion	0.0714	0.0711	0.0
9	democracy_importance	0.0527	0.0523	0.0
8	gender_equality	0.0454	0.0451	0.0
6	life_satisfaction	0.0419	0.0416	0.0
3	importance_friends	0.0346	0.0343	0.0
4	national_pride	0.0338	0.0334	0.0
7	political_orientation	0.0157	0.0154	0.0
1	importance_family	0.0059	0.0056	0.0

In Table 7 above, the MVLR results demonstrate that predictive performance varies across outcomes. With the highest explanatory power of $R^2 = 0.1$, importance of work is followed by trust in people and importance of religion, with R^2 of 0.085 and 0.071, respectively. Variables such as importance of democracy and gender equality have moderate explanatory power, whereas family importance has low explanatory power. The huge differences highlight that some social values are more closely tied to demographic characteristics than others.

The cross-validation results closely match the in-sample estimates, indicating that the models generalize well without overfitting. Notwithstanding, the overall R^2 values remain relatively low, giving the conclusion that demographic variables can only explain a limited portion of the variation in values.

Table 8: Cross-validated MVLR performance

	response	cv_rmse	cv_r2	obs_pred_corr
2	importance_work	0.8323	0.1002	0.3165
5	trust_people	0.4373	0.0844	0.2906
0	importance_religion	1.0984	0.0704	0.2653
9	democracy_importance	2.0995	0.0519	0.2277
8	gender_equality	2.2723	0.0447	0.2115
6	life_satisfaction	2.1070	0.0413	0.2033
3	importance_friends	0.9465	0.0337	0.1837
4	national_pride	0.9139	0.0328	0.1811
7	political_orientation	2.3968	0.0143	0.1199
1	importance_family	0.3749	0.0041	0.0650

The impact of the missingness of the variable political orientation is addressed in the sensitivity analysis. When this variable is removed, the sample size increases to 16,537 with all 10 countries retained. Correspondingly, the first canonical correlation decreases to 0.3759, and the mean cross-validated R^2 also declines. Political orientation, therefore, provides meaningful insight into the relationship between demographics and values. However, the overall pattern remains similar, with the main conclusions being robust and not easily driven by missingness in a single variable.

Conclusion

In conclusion, this report looks at whether demographic characteristics are systematically related to social values in the World Values Survey. It is shown that a meaningful but only moderate relationship exists. PCA indicates that the data are multidimensional, and CCA identifies a dominant first canonical relationship linking demographic variables to value variables. Furthermore, MVLR results show that some outcomes are more predictable than others, although overall explanatory power remains limited.

Table 9: Sensitivity summary

	analysis	n	countries	first_canonical_correlation	mean_cv_r2
0	main_complete_case	11919	9	0.4779	0.0478
1	drop_political_orientation	16534	10	0.3795	0.0316

Even more, the sensitivity analysis confirms that these findings are not merely due to missingness in political orientation. In summary, demographic variables provide useful yet incomplete explanations of social values. The relationship could be treated as real but not deterministic, which reflects the complexity of social values in survey data.

Reference

Haerpfer, Christian W., Ronald F. Inglehart, Alejandro Moreno, Christian Welzel, Kseniya Kizilova, Jaime Diez-Medrano, Marta Lagos, et al. 2022. World Values Survey: Round Seven - Country-Pooled Datafile Version 5.0. Madrid, Spain & Vienna, Austria: JD Systems Institute & WVSA Secretariat. https://doi.org/10.14281/18241.18